Sparse target packing and arithmetic separation for Diophantine consistency

Explicit equations for the absence of a ZFC refutation

Research prepared for Vladimir Reshetnikov and the ProveIt project

23 October 2026

Abstract

We construct explicit improvements to three measures of simplicity for a Diophantine equation whose lack of solutions is provably equivalent, in ZFC, to syntactic consistency of ZFC. The principal unrestricted-integer equation has ten variables, exact total degree 54 768 104, and a division-free circuit of 709 binary arithmetic operations. A companion has nine strictly positive variables, exact degree 33 332 620, and 682 operations. Independently, a squared-modulus Pell congruence reduces the fifteen-positive-variable operation record from 67 to 66, at exact degree 157, and gives 72-operation/degree-80 and 74-operation/degree-68 alternatives.

Three ingredients supply the improvement: coefficient extraction at positions selected using the actual quadratic supports; translation of positivity inside the packed word; and a square-root separator proved by divisibility and parity instead of large magnitude bounds. Every exponent and coefficient is fixed. The accompanying archive contains literal circuits, coefficient payloads, deterministic builders, independent finite certificates, source pins, and detailed dependency proofs. The new equivalences are ordinary mathematical proofs formalizable in ZFC. We do not claim a new end-to-end Lean/Rocq certificate, a proof of consistency, or global minimality.

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1 Results, domains, and the meaning of simplicity

The reviewed repository snapshot is

`124c07c0b77a9e20e84676514d8b1ff78536ea3f`.

It contains the newly placed wave-6 manuscripts and the active formal development in `Computability/DiophantineConsistency`. Comparisons refer to that snapshot, rather than the superseded wave-5 summary alone.

Theorem 1.1 (Explicit consistency equations)**.** *There are the explicitly specified fixed integer polynomials F_{10}, F_9, and P_{66} in this report and archive for which ZFC proves*

$$\begin{aligned}
\neg\exists \mathbf{x} \in \mathbb{Z}^{10}\ F_{10}(\mathbf{x}) = 0 &\iff \mathrm{Con}(\mathsf{ZFC}),\\
\neg\exists \mathbf{x} \in \mathbb{Z}^{9}_{>0}\ F_{9}(\mathbf{x}) = 0 &\iff \mathrm{Con}(\mathsf{ZFC}),\\
\neg\exists \mathbf{x} \in \mathbb{Z}^{15}_{>0}\ P_{66}(\mathbf{x}) = 0 &\iff \mathrm{Con}(\mathsf{ZFC}).
\end{aligned}$$

Their exact degrees and delivered circuit sizes are in Table 1. The theorem uses the inherited K215 calibration and ordinary native-compiler correctness theorem. The ten-integer branch also uses Sun's Lucas/interpolation theorems, and the nine-positive branch uses the Jones ratio bridge. These dependencies are identified explicitly below.

Polynomial	Witness domain	Exact degree	Binary operations
F_{10}	$\mathbb{Z}^{10}$	54 768 104	709
F_9	$\mathbb{Z}^{9}_{>0}$	33 332 620	682
$F_{9,\mathbb{N}}$	$\mathbb{N}^{9}$	33 332 620	685
P_{66}	$\mathbb{Z}^{15}_{>0}$	157	66
P_{68}	$\mathbb{Z}^{15}_{>0}$	113	68
P_{72}	$\mathbb{Z}^{15}_{>0}$	80	72
P_{74}	$\mathbb{Z}^{15}_{>0}$	68	74

Table 1: New explicit presentations; $\mathbb{N} = \{0, 1, 2, \ldots\}$. Subscripts 66, 68, 72, and 74 count operations, not variables.

We count every binary addition, subtraction, and multiplication, including multiplication by a fixed numeral. Reuse is free. A fixed integer coefficient is a leaf; its finite construction and binary value are supplied separately. Fixed powers are expanded into multiplication schedules. There are no division, root, sign, divisibility, or variable-exponentiation gates.

The closest low-variable baseline is package 40: ten unrestricted integers, degree 237 520 456, and 782 operations. Package 38 has a different ten-integer tradeoff, 773 operations at degree 449 607 016. Package 45 gives nine positive variables at degree 129 875 980 and 788 operations. The new ten-integer equation improves both of these ten-integer comparators. The nine-positive equation improves both degree and operations against package 45. Degree drops by factors approximately 4.337 and 3.896, respectively. These are concrete improvements to the current repository; they do not settle a general minimum-variable or minimum-degree problem.

The 66-operation equation is a different tradeoff. It raises the old P67 degree from 155 to 157. Its 72/80 and 74/68 companions improve the checked 73/80 and 75/68 operation counts without changing those degrees. The new circuits have not themselves been submitted to a proof-assistant kernel.

2 Semantic starting point and fixed coefficients

2.1 The exact logical statement

Here Con(ZFC) means absence of a finite syntactic derivation of falsity from the specified ZFC axiom predicate. The repository's literal machine K215, on blank tape, is calibrated to this predicate:

$$K215 \text{ halts on blank input} \quad \longleftrightarrow \quad \text{ZFC has a finite refutation.} \tag{1}$$

The formal endpoint is `BusyBeaver.ZFC.KEndpoint215.blank_halts_iff_zfc_refutation`. The explicit equations use the ordinary native-window/Pell compiler theorem. No consistency assumption or search for a refutation enters their coefficient construction.

For the low-variable constructions retain the *complete* package-31 tuple

$$(\beta, K_0, M_C, M_F), \qquad \beta = B_{\mathrm{nat}} - 1, \quad K_0 = D_C + B_{\mathrm{nat}} D_R,$$

where $B_{\mathrm{nat}} = 2^{1064921}$ and $M_F = M_{F,\mathrm{native}} + B_{\mathrm{nat}} - 1$ is already shifted. These are fixed integers, not quantified variables. Their binary payloads, deterministic generator, 943-tile table, and literal K215 transition table are bundled. Their respective bit lengths are

$$1064921, \quad 1070004, \quad 1064921, \quad 1064922.$$

The short equation in Section 9 uses instead the *whole* package-36 tuple with $B_{\mathrm{nat}} = 2^{1012115}$. The two tuples are never spliced or silently identified.

2.2 The quadratic parent

The 31 positive parent coordinates, in their data order, are

$$J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma,$$
$$N, E, P, c, a, X, D, C_2, L, F_2, V, g, I, B_{\mathrm{aux}}, \Delta.$$

Write

$$q = \beta J + 1, \qquad k = \eta + \zeta, \qquad H_p = 4a + 3, \qquad C_p = q - F - \alpha.$$

Table 2 gives the exact system. The last column anticipates our new extraction positions.

Table 2: Unchanged quadratic parent and new targets.

Row	Equation	Target
1	$L = q^2 - 1$	714
2	$N = q(L + 1)$	6707
3	$X = wq$	6262
4	$a = E + sN$	634
5	$E = X(a - E)$	6148
6	$c = k(a - E) + \eta$	5133
7	$P = E(c - \eta) + k$	4099
8	$\tau^2 = P(P - k) + 1$	7243
9	$D = X + ac + \gamma H_p$	1024
10	$\Delta = a^2 + H_p$	2665
11	$C_2 = c^2$	4617
12	$D^2 = \Delta C_2 + 1$	5288

Row	Equation	Target
13	$k = jc - V + 1 + hE$	2271
14	$jc - V = (q - F)(N - q) - C_pL + (M_C + qM_F)J$	3114
15	$(K_0 + w)C_p + q - F - z(q - 1) = 1$	1023
16	$F_2 = f^2$	5402
17	$V = of - c$	5193
18	$g = y^2 - V^2$	2781
19	$I = iC_2$	1906
20	$B_{\mathrm{aux}} = I^2$	6470
21	$B_{\mathrm{aux}} = \Delta(F_2 - 1)$	6168
22	$y^2 - B_{\mathrm{aux}}g = 1$	4539

After expanding the affine abbreviations, all residuals are quadratic. Let r_j be left side minus right side except that row 8 is negated. The inherited parent theorem is

$$\exists \mathbf{p} \in \mathbb{Z}_{>0}^{31} \ \bigwedge_{j=1}^{22} r_j(\mathbf{p}) = 0 \quad \longleftrightarrow \quad \neg\,\mathrm{Con}(\mathsf{ZFC}). \tag{2}$$

Negating a residual preserves this theorem. Row 8 now has constant term $+1$, useful for positivity of the packed polynomial. The all-ones tuple is not a parent zero: row 5 has residual $1 - 1(1 - 1) = 1$ there.

The exact parent circuit and a separate sparse description are included. Our checker proves their equality over the integer polynomial ring, treating all native ports as independent symbols. This finite identity does not replace the inherited semantic calibration.

3 Coefficient extraction using actual supports

Let $r_1, \dots, r_m \in \mathbb{Z}[p_1, \dots, p_n]$ be nonzero polynomials of degree at most two. Choose positions $w_0 = 0, w_1, \dots, w_n > 0$ with all unordered pair sums $w_i + w_j$, including repetitions, distinct. Define

$$\mathcal{S} = \{w_i + w_j : 0 \le i \le j \le n\}, \qquad e(\alpha) = \sum_i \alpha_i w_i.$$

Write $r_j = \sum_\alpha a_{j,\alpha} p^\alpha$ and $\mathcal{S}_j = \{e(\alpha) : a_{j,\alpha} \neq 0\}$.

Lemma 3.1 (Selective target extraction)**.** *Suppose distinct targets $E_1, \dots, E_m$ satisfy*

$$E_j \ge \max \mathcal{S}_j, \tag{3}$$

$$(E_k - E_j + \mathcal{S}_j) \cap \mathcal{S} = \varnothing \quad (j \neq k). \tag{4}$$

Set

$$\mathcal{D}(Z) = \sum_{j,\alpha} (2 - |\alpha|)! \prod_i \alpha_i!\, a_{j,\alpha} Z^{E_j - e(\alpha)}. \tag{5}$$

Then the exact polynomial identity

$$[Z^{E_j}] \left(1 + \sum_i p_i Z^{w_i}\right)^2 \mathcal{D}(Z) = 2 r_j(\mathbf{p}) \tag{6}$$

holds for every j. If every coefficient of each r_j has magnitude at most H, every coefficient of $\mathcal{D}$ has magnitude at most $2H$. For $p_i \ge 0$, every product coefficient has magnitude at most $2H(1 + \sum_i p_i)^2$.

Proof. Pad monomials of degree below two with zero-weight letters. Pair-sum uniqueness makes the coefficient of $Z^{e(\alpha)}$ in the square

$$\frac{2!}{(2-|\alpha|)!\prod_i \alpha_i!}\, p^\alpha.$$

It cancels the multiplier in (5) at its own target. A contribution from row j to a different E_k would require $E_k - E_j + e(\alpha) \in \mathcal{S}$, excluded by (4).

No two terms from different rows have equal reversal exponent, since $E_j - e(\alpha) = E_k - e(\beta)$ implies the same forbidden membership. Within a row, uniqueness also prevents collisions. Thus each coefficient of $\mathcal{D}$ is one weighted coefficient, with factorial multiplier at most two. The square has nonnegative coefficients summing to $(1+\sum p_i)^2$; convolution gives the last bound. □

The full set $\mathcal{S}$ is indispensable: the packed square includes pair monomials absent from every residual. The smaller actual support $\mathcal{S}_j$ is legitimate only on the source side. Uniform target spacing greater than $2\max w_i$ is sufficient but unnecessary. We allow whole support bands to overlap, forbidding contamination only at the targets actually read.

Use the same 32-element pair-sum set as package 40:

$$\begin{aligned}(w_0,\dots,w_{31}) = (&0, 10, 35, 55, 62, 105, 144, 145, 159, 168, 232,\\ &245, 306, 323, 325, 353, 371, 374, 382, 494, 499, 525,\\ &566, 578, 610, 647, 681, 685, 741, 777, 783, 799).\end{aligned}$$

The 528 pair sums are distinct. Their finite certificate is independent of the search that found the set; historically it comes from a Bose–Chowla construction. With Table 2,

$$t = 799, \qquad E_{\max} = 7243, \qquad M = E_{\max} + 2t = 8841.$$

The independent audit checks all 1974 off-diagonal conditions and $94 \cdot 528 = 49\,632$ dense convolution pairs. It recovers all 22 residuals over symbolic ports. The coefficient of Z^{7243} in $\mathcal{D}(Z)$ is exactly 2.

The data order places negated row 8 last; that order is separate from the original row labels in the table. The target map is explicit. The new positions were found by allowing insertion before existing targets. The search has no mathematical authority: the finite conditions above certify the result.

4 Translate positivity in the packed word

A radix digit is nonnegative, while the parent coordinates are positive. Expanding every substitution $p_i = n_i + 1$ in every residual creates unnecessary records. Instead put

$$\ell(Z) = \sum_i Z^{w_i}, \qquad g(Z) = \sum_i n_i Z^{w_i}.$$

Then

$$1 + g(Z) + \ell(Z) = 1 + \sum_i (n_i + 1) Z^{w_i},$$

and Lemma 3.1 gives

$$[Z^{E_j}](1 + g(Z) + \ell(Z))^2 \mathcal{D}(Z) = 2r_j(\mathbf{n} + \mathbf{1}). \tag{7}$$

This represents every positive parent tuple. It does not restrict coordinates to squares or a sparse subsequence.

There are 94 weighted records, or 99 terms after expanding port polynomials, instead of 136 records and 161 terms in the shifted baseline. The register ℓ is already required by the low digit mask; inserting

it in the square costs one addition. The finalizer's variable g itself is unchanged; only the expression inside this square is translated.

Set the fixed bounds

$$H = 8(1 + \beta + K_0 + M_C + M_F)^2, \qquad \beta_S = 2^{2140030}.$$

Raw coefficients have port degree at most two and coefficient ℓ^1 norm at most four, so eight is conservative. Exact integer checks give

$$\beta_S > 4 \cdot 33^2 H, \qquad \beta_S > 36H + 2. \tag{8}$$

The even exponent makes β_S a square, a power of two, and 1 modulo three. The nine nonliteral recipes in $\mathcal{D}$ and β_S are supplied as actual fixed signed-magnitude binary values, not unspecified indices.

5 The two-variable source for unrestricted integers

Letters here through Section 7 refer to final coordinates, not the parent. Start with $f, g \in \mathbb{Z}$ and define polynomial registers

$$b = 1 + 3f, \qquad \mathcal{B} = \beta_S b^2, \qquad \ell = \sum_i \mathcal{B}^{w_i},$$

$$N_0 = \mathcal{B}^{t+1}, \qquad N_1 = 4\mathcal{B}^{M+1}, \qquad N = N_0 N_1, \tag{9}$$

$$M_c = N_0 - 1 - (b-1)\ell, \qquad Q_c = (\mathcal{B} - 2)\sum_j \mathcal{B}^{E_j}, \qquad T_c = M_c + Q_c N_0,$$

$$J_c = 2(1 + g + \ell)^2 \mathcal{D}(\mathcal{B}) + \mathcal{B}\sum_{i=0}^{M} \mathcal{B}^i, \qquad S_c = g + J_c N_0, \tag{10}$$

$$R_c = (S_c + T_c + 1)N + T_c + 1, \qquad X = (N-1)R_c, \qquad Y = N^2,$$

$$\mathscr{C} = b\mathcal{B}^t. \tag{11}$$

Here $N = 4\mathcal{B}^{9642}$. All sums and powers have fixed indices.

Proposition 5.1 (Source interface)**.** *(i) A positive parent zero supplies, for arbitrarily large $f > 0$ with b a square and a power of two, an integer g with $b \le g < \mathscr{C}$ and $Y \mid \binom{2X}{X}$.*

(ii) If $f \neq 0$, b is a square and a power of two, $0 \le g < 2\mathscr{C}$, and $Y \mid \binom{2X}{X}$, the parent has a positive zero.

(iii) Assuming only $f \neq 0$, b square, and $0 \le g < 2\mathscr{C}$ gives $0 \le S_c, T_c < N$, $3 \mid X$, $X > 3b$, and $Y > \max\{b, 256\}$.

Proof. First establish broad ranges without assuming binary encoding. A square $b = 1 + 3f$ with $f \neq 0$ has $b \ge 4$ and $f > 0$. The leading coefficient of $\mathcal{D}$ is 2; all others have magnitude at most $2H$. Since $\mathcal{B} > 2H + 1$, $\mathcal{D}(\mathcal{B}) > 0$. The radix digits of M_c are $\mathcal{B} - b$ at selected positions and $\mathcal{B} - 1$ elsewhere, so $0 \le M_c < N_0$. Also $0 \le Q_c < \mathcal{B}^{7244} < N_1$, whence $0 \le T_c < N$.

For the entire interval $0 \le g < 2b\mathcal{B}^t$, integrality and $\ell < 2\mathcal{B}^t$ give $1 + g + \ell < 3b\mathcal{B}^t$ and $g < N_0$. Using (8),

$$0 < J_c < \frac{36Hb^2 + \mathcal{B}}{\mathcal{B} - 1}\mathcal{B}^{M+1} < 2\mathcal{B}^{M+1} < N_1.$$

Thus $0 \le S_c < N$. Moreover $\beta_S \equiv b \equiv \mathcal{B} \equiv N \equiv 1 \pmod 3$ gives $3 \mid X$. Since $R_c \ge N + 1 > \mathcal{B} > b$ and $N \ge 4$, $X > 3b$. The fixed radix gives the remaining bound on Y.

Now let b be a power of two. The low mask has no binary overlap with g exactly when $g = \sum_i n_i \mathcal{B}^{w_i}$ with $0 \le n_i < b$. Put $p_i = n_i + 1$. Each coefficient a_s in (7) obeys

$$|a_s| \le 2H(1 + \textstyle\sum_i p_i)^2 < 2H(33b)^2 < \mathcal{B}/2.$$

Thus the radix digits of

$$K_c = (1 + g + \ell)^2 \mathcal{D}(\mathcal{B}) + (\mathcal{B}/2) \sum_{s=0}^{M} \mathcal{B}^s$$

are exactly $\mathcal{B}/2 + a_s$, strictly between zero and $\mathcal{B}$. This is a proof abbreviation; the circuit evaluates $J_c = 2K_c$ without division. At target E_j the digit is $\mathcal{B}/2 + 2r_j(\mathbf{p})$. For $0 < d < \mathcal{B}$, no overlap with $\mathcal{B}/2 - 1$ holds precisely when $d = \mathcal{B}/2$. The high mask Q_c, after a common shift by one binary bit, applies this test at every target. Low and high blocks are disjoint, so

$$\text{no carries in } S_c + T_c \iff g \text{ encodes digits and } r_j(\mathbf{n} + \mathbf{1}) = 0 \text{ for all } j.$$

Lemma 5.2 below identifies this with $N^2 \mid \binom{2X}{X}$, proving (ii).

Conversely, given a parent zero, set $n_i = p_i - 1$ and choose arbitrarily large $b = 2^{2q} > \max_i n_i$, so $f = (b-1)/3$ is positive. The all-ones tuple is not a parent zero, so some $n_i > 0$ and $g = \sum n_i \mathcal{B}^{w_i} \ge \mathcal{B} \ge b$. A geometric-series bound gives $g < b\mathcal{B}^t$. Both masks pass, giving (i). □

Lemma 5.2 (Central binomial carry identity)**.** *For $N = 2^e$, $0 \le S, T < N$, and $X = (N-1)((S+T+1)N + T + 1)$,*

$$N^2 \mid \binom{2X}{X} \quad \iff \quad \textit{binary addition of } S, T \textit{ has no carries.}$$

Proof. The exact base-N expression is $X = (S+T)N^2 + (N-1-S)N + (N-1-T)$. Hence $\mathrm{pc}(X) = 2e + \mathrm{pc}(S+T) - \mathrm{pc}(S) - \mathrm{pc}(T) = 2e - \tau_2(S,T)$, where τ_2 counts carries with multiplicity. The factorial valuation formula gives $\nu_2\binom{2X}{X} = \mathrm{pc}(X)$, proving the assertion. □

6 The explicit ten-integer polynomial

The ten unknowns are

$$(f, g, h, k, l, w, x, y, z, m) \in \mathbb{Z}^{10}.$$

Use the source quantities $b, \mathcal{B}, \ell, N, X, Y, \mathrm{cap}$ from Section 5, writing $\mathrm{cap} := \mathscr{C}$. Letters introduced below are polynomial abbreviations, not additional unknowns. In particular, g remains the original unknown in every expression below; the successor translation is confined to the source expression $(1 + g + \ell)^2 \mathcal{D}(\mathcal{B})$.

Define the following registers:

$$\begin{aligned}
&L = lY, && U = 2LX, && V = 4gwY, && W = bw,\\
&A = U(V+1), && B_1 = 2X + 1, && C = B_1 + (A-2)h, &&\\
&P = U^2 V, && K = X + 1 + k(P-2), && S = 2A - 5, &&\\
&T = 3WC - 2(W^2 - 1), && r_1 = (P^2 - 4)K^2 + 4. &&&&
\end{aligned}$$

The range margin and positivity register are

$$\begin{aligned}
\Delta_b &= 8\,\mathrm{cap}^3 g K^2 - g^2\big(32(C - KL)^2 \mathrm{cap}^3 + g^2 K^2\big),\\
O &= f^2 l^2 \Delta_b, \qquad \xi = zx.
\end{aligned}$$

The modified Lucas certificate is

$$\begin{aligned} D &= (A^2-4)C^2+4, & E &= C^2D\xi,\\ F &= 4(A^2-4)E^2+1, & G_* &= 2(A+2)(A-2)^2E^2-1,\\ H_* &= C+(B_1-C)F+16EFy, & I_* &= (G_*^2-1)H_*^2+1,\\ r_2 &= DFI_*. \end{aligned}$$

Set

$$\begin{aligned} r_0 &= b(1+(3O-4)b^2z^2),\\ \Sigma &= \xi S^5, \qquad r = T+Sy,\\ \Theta &= (r^3-rS^2-S^3)(r^2+23S^2),\\ Z_{\mathrm{n}} &= \Sigma m-\Theta,\\ a_0 &= \Sigma^2 r_0, \qquad a_1 = \Sigma^2 b^2 r_1, \qquad a_2 = \Sigma^2 b^2 K^2 r_2. \end{aligned}$$

Finally put

$$\boxed{\begin{aligned} V_{\mathrm{n}} &= Z_{\mathrm{n}}^2+a_2-a_0-a_1,\\ W_{\mathrm{n}} &= V_{\mathrm{n}}^2+4(Z_{\mathrm{n}}^2a_2-a_0a_1),\\ F_{10} &= W_{\mathrm{n}}^2-16V_{\mathrm{n}}^2Z_{\mathrm{n}}^2a_2. \end{aligned}} \tag{12}$$

The requested unrestricted-integer equation is $F_{10}=0$. The complete expanded circuit, including fixed-power schedules, is `data/polynomial.json`. Thus this display specifies a single polynomial, despite its readable sequence of abbreviations.

6.1 The norm identity

For indeterminates t, a_0, a_1, a_2, let

$$\mathcal{N}(t,a_0,a_1,a_2) = \big((t^2+a_2-a_0-a_1)^2+4(t^2a_2-a_0a_1)\big)^2-16(t^2+a_2-a_0-a_1)^2t^2a_2.$$

Multiplication of conjugate factors proves the exact identity

$$\mathcal{N}(t,a_0,a_1,a_2) = \prod_{\varepsilon\in\{-1,1\}^3}\big(t+\varepsilon_0\sqrt{a_0}+\varepsilon_1\sqrt{a_1}+\varepsilon_2\sqrt{a_2}\big). \tag{13}$$

It is valid over the complex numbers, including zero radicands and coincident squareclasses. All radicals cancel; the result is an integer polynomial monic of degree eight in t. The audit separately expands both sides to the same 35-term polynomial.

The conceptual weights in this construction are $(1, b, bK)$. The previous package-40 weights were $(1, r_0, r_0(P^2+K^2+3))$. The smaller weights are valid because of arithmetic information about these particular radicands. They are not a universal square-test identity for arbitrary inputs.

7 An arithmetic square-root separator

Lemma 7.1 (Coprimality separates one squareclass)**.** *Let $b, J_0, A_0, A_1, K \in \mathbb{Z}$, with $|b|>1$ and $\gcd(b,J_0)=1$. If a signed sum*

$$\sqrt{bJ_0} \pm b\sqrt{A_0} \pm bK\sqrt{A_1}$$

is rational, then bJ_0 is a nonnegative integer square.

Proof. If bJ_0 is not such a square, write $bJ_0 = da^2$ with $a > 0$ and $d \neq 1$ a signed squarefree integer. Distinct signed squarefree radical classes are linearly independent over $\mathbb{Q}$: they are distinct characters in the multiquadratic extension generated by the radicals. The coefficient of $\sqrt{d}$ must therefore cancel. Every contribution from the last two terms is an integer multiple of b, so $b \mid a$. Then $b^2 \mid bJ_0$, hence $b \mid J_0$, contradicting coprimality. This argument also covers negative d. □

Lemma 7.2 (A Pell remainder prevents two-root cancellation)**.** *Suppose $P, K, R \in \mathbb{Z}$, $K \neq 0$, and $4 \mid R$. Put $A_0 = (P^2-4)K^2+4$. If a signed sum $\sqrt{A_0} \pm K\sqrt{R}$ is rational, then both A_0 and R are nonnegative integer squares.*

Proof. Call the signed rational sum q. If $q \neq 0$, squaring gives a rational expression for $\sqrt{A_0}$ with denominator $2q$, and then one for $\sqrt{R}$. A rational square root of an integer is an integer. If $q = 0$, then $A_0 = K^2R$, so $K^2 \mid 4$. For $|K| = 1$ this gives $A_0 = R = P^2$. For $|K| = 2$ it gives $R = P^2 - 3$, which is 1 or 2 modulo 4, contradicting $4 \mid R$. □

Proposition 7.3 (All square predicates are recovered)**.** *Every integer zero of F_{10} satisfies*

$$\Sigma \neq 0, \quad \Sigma \mid \Theta, \quad S \mid T, \quad b \text{ is a positive square}, \quad O > 0, \quad r_1, r_2 \text{ are nonnegative squares.}$$

Proof. The integer $S = 2A - 5$ is odd and nonzero. The polynomial

$$\pi(v) = (v^3 - v - 1)(v^2 + 23)$$

has no rational root. Consequently $\Theta = S^5\pi((T + Sy)/S) \neq 0$ for every integer assignment. At $\Sigma = 0$ equation (12) is Θ^8. Thus a zero has $\Sigma \neq 0$, whence $\xi, z, x \neq 0$.

Divide the norm equation by Σ^8. The rational number $m - \Theta/\Sigma$ is a root of a monic integer polynomial with radicands $r_0, b^2r_1, b^2K^2r_2$. The rational-root theorem gives $\Sigma \mid \Theta$. The product identity also gives a rational signed sum

$$\varepsilon_0\sqrt{r_0} + \varepsilon_1 b\sqrt{r_1} + \varepsilon_2 bK\sqrt{r_2}. \tag{14}$$

Signs of integer weights may be absorbed into the independent signs.

Write $J_0 = 1 + (3O - 4)b^2z^2$. Then $\gcd(b, J_0) = 1$. Since $b = 1 + 3f$, it is nonzero and is never -1. If $|b| > 1$, Lemma 7.1 implies that $r_0 = bJ_0$ is a square.

We handle the remaining possibility $b = 1$ before assuming positivity. It gives $f = O = 0$ and $r_0 = 1 - 4z^2$, an odd negative integer. Both r_1 and r_2 are divisible by four: P, A are even and $D = (A^2 - 4)C^2 + 4$. Write $r_0 = da^2$ with signed squarefree $d \neq 1$. Both d and a are odd. A radical from either of the last two terms of (14) in this same squareclass has an even integer coefficient. Its cancellation would require an odd integer plus even integers to vanish. Thus $b = 1$ is impossible.

Also $r_0 \neq 0$. Indeed, $J_0 = 0$ would imply $(3O-4)b^2z^2 = -1$, hence $|b| = |z| = 1$ and $O = 1$. But then $b = 1, f = 0, O = 0$, a contradiction. Thus r_0 is a positive square. Unique factorization and coprimality show that b, J_0 are signed squares with the same sign. A negative nonzero square is never 1 modulo 3, so $b, J_0 > 0$. If $O \leq 0$, then $J_0 \leq 1 - 4(bz)^2 < 0$, a contradiction. Therefore $O > 0$, $f, l \neq 0$, and $\Delta_b > 0$.

Since $\mathrm{cap} = b\mathcal{B}^t > 0$, the case $K = 0$ would give $\Delta_b = -32g^2C^2\mathrm{cap}^3 \leq 0$. Thus $K \neq 0$. Subtract the integral first root in (14) and divide by b. Lemma 7.2, with $R = r_2$, proves that r_1, r_2 are squares. No source size bound, preliminary positivity of g, or Lucas recovery was used in these steps.

It remains to justify $S \mid T$. If not, choose a prime with $\alpha = \nu_p(S) > \beta = \nu_p(T)$. In the two homogeneous factors of Θ, the terms r^3 and r^2 have uniquely smallest valuations, respectively 3β and 2β, because $\nu_p(r) = \nu_p(T + Sy) = \beta$. It follows that $\nu_p(\Theta) = 5\beta < 5\alpha$, contradicting $S^5 \mid \Theta$. □

7.1 The inherited Lucas bridge and its precise use

The following are the existing Sun/package-31 bridge statements, used in the parametric form justified in package 40 [2, 6]. Let $u_0(A) = 0, u_1(A) = 1$, and $u_{n+1}(A) = Au_n(A) - u_{n-1}(A)$. For a source satisfying Section 5:

1. A positive parent zero supplies positive bridge witnesses with $A > 2$ even, $B_1 > 0$ odd,
$$b \text{ square}, \quad b \le g < \text{cap}, \quad C = u_{B_1}(A), \quad r_1 \text{ square}, \quad S \mid T, \quad O > 0.$$
2. For arbitrary signed inputs, the conditions
$$b \text{ square}, \quad O > 0, \quad \xi \neq 0, \quad r_1, r_2 \text{ squares}, \quad S \mid T$$
imply that b is a power of two, $0 < g < 2\,\text{cap}$, and $Y \mid \binom{2X}{X}$.
3. For $A > 2$ even, $B_1 > 0$ odd, and $C = u_{B_1}(A)$, every positive integer d admits $\xi > 0$ divisible by d such that each residue class modulo ξ contains a y making r_2 a square.

These are inherited arithmetic theorems, not outputs of a finite audit. The full original proof and source-dependent hypotheses are included in `upstream/PACKAGE40_REPORT.tex` and `upstream/SIGNED10_PROOF.md`.

The new source preserves every premise used in those parametric proofs: $b = 1 + 3f$, the forward and reverse central-binomial conditions, the pre-decoding bounds $3 \mid X$, $X > 3b$, and $Y > \max(b, 256)$, and
$$\text{cap} = 2^{1709883970} b^{1599}.$$
The fixed exponent of two is even and the exponent of b is odd, as required by the inherited positive bridge. The new target schedule changes X, Y and their degrees, but not this cap.

Theorem 7.4 (Ten-integer equivalence)**.** *The polynomial of* (12) *has a zero in* $\mathbb{Z}^{10}$ *if and only if the positive quadratic parent has a zero. Consequently its nonsolvability is provably equivalent in ZFC to* $\text{Con}(\mathsf{ZFC})$.

Proof. For soundness, Proposition 7.3 gives all hypotheses of the reverse bridge. Source soundness then decodes a positive parent zero. For clarity, $O > 0$ itself implies $g > 0$ and
$$0 < g < 2\,\text{cap}, \qquad 4(C - KL)^2 < K^2.$$
Indeed $g \le 0$ makes $\Delta_b \le 0$; with $g > 0$ its positivity yields $g^3 < 8\,\text{cap}^3$ and $4g(C - KL)^2 < K^2$. This supplies the broad source range before binary base recovery. There is no assumption that g already encodes digits.

Conversely, the forward bridge supplies the witnesses in item 1. The positive integer Δ_b is a multiple of g, so $O = f^2 l^2 \Delta_b \ge g \ge b \ge 4$. Then $d = 3O - 4 > 0$ is 2 modulo 3, hence not a square. A positive Pell unit for d, raised to a suitable finite order modulo b, gives $u^2 - dv^2 = 1$ with $v > 0$ and $b \mid v$. Taking $z = v/b > 0$ makes $r_0 = bu^2$ a square.

The polynomial π has a root modulo every positive integer. For completeness, its cubic factor has discriminant -23. At an odd prime other than 23, either the cubic has a simple root, or its irreducibility makes the discriminant a square and the quadratic has a simple root. At 23, the cubic has the simple root 3. At 2, $-23 \equiv 1 \pmod 8$ is a 2-adic square. Hensel lifting and the Chinese remainder theorem give the assertion.

Now choose $\xi > 0$ divisible by z using item 3, and put $x = \xi/z$. Write $T = Sq$. Choose a root c of π modulo ξ and prescribe $y \equiv c - q \pmod \xi$. Interpolation supplies such a y with r_2 square, while $\Theta = S^5 \pi(q + y)$ is divisible by $\Sigma = \xi S^5$. All three weighted square roots are integers. Choose m so that one factor of (13) vanishes. This proves completeness. Finally apply the inherited parent equivalence (2). □

7.2 Exact cost and degree

The delivered circuit contains 189 additions, 25 subtractions, and 495 multiplications, totaling 709 live binary operations. It shares $\Sigma^2 b^2$, K^2, and $Z_{\mathrm{n}}^2 a_2$. The displayed factorization of W_{n} saves a scalar multiplication. The conceptual product bK need not be computed as a separate gate.

Degree propagation gives the total-degree upper bound 54 768 104. Specialize all ten variables to one indeterminate T and compute the leading coefficient at that upper bound using the actual fixed coefficient tuple. Modulo 1 000 000 007 it is

$$323\,596\,601 \neq 0.$$

This proves that the bound is attained over $\mathbb{Z}$, hence that it is the exact total degree. This is exact coefficient arithmetic, not a probabilistic identity test. Independent literal-formula replay also checks nonzero leaders modulo two further primes.

8 The nine-positive-variable companion

This branch combines the same new quadratic source with the existing Jones ratio finalizer in package 45 [3]. Its raw positive parent is the package-40 parent in Table 2; package 45 used an affine-shifted parent. The source interface, rather than a claim of pointwise equality between those parents, justifies the composition.

8.1 Source and domain

Initially take mixed coordinates

$$\epsilon, \phi, n \in \mathbb{N}, \qquad g, h, s, w, i, j \in \mathbb{Z}_{>0}.$$

Define the fixed constants

$$M_{\mathrm{coef}} = 2^{2134927}, \qquad b_{\mathrm{floor}} = 2^{2134939} = 4096 M_{\mathrm{coef}},$$

and $b = b_{\mathrm{floor}} + \epsilon$, $\mathcal{B} = b^3$. Every weighted coefficient of the same $\mathcal{D}$ has absolute value less than M_{coef}; the exact recipes and bit-length audit certify this bound. With the same $t = 799$, $E_{\max} = 7243$, and $M = 8841$, set

$$\ell = \sum_{i=1}^{31} \mathcal{B}^{w_i}, \qquad N_0 = \mathcal{B}^{t+1}, \qquad N_1 = 4\mathcal{B}^{M+1}, \quad N = N_0 N_1,$$

$$M_c = N_0 - 1 - (b-1)\ell, \qquad Q_c = (\mathcal{B}-2)\sum_{j=1}^{22} \mathcal{B}^{E_j},$$

$$T_c = M_c + Q_c N_0,$$

$$J_c = 2(1+g+\ell)^2 \mathcal{D}(\mathcal{B}) + \mathcal{B}\sum_{a=0}^{M} \mathcal{B}^a, \qquad S_c = g + J_c N_0,$$

$$R = (N-1)\big((S_c + T_c + 1)N + T_c + 1\big), \quad G_{\mathrm{cap}} = 32 b \mathcal{B}^t.$$

These formulas have unconditional positivity on the mixed domain. In particular $\mathcal{D}(\mathcal{B}) > 0$ because its leading coefficient is 2 and $\mathcal{B} - 1 > M_{\mathrm{coef}}$. Hence $R, N \geq 8$, $b \leq R, N$, and $b, G_{\mathrm{cap}} > 0$.

The broad range needed for soundness is $0 < g < 4b\mathcal{B}^t$. Integrality and $\ell < 2\mathcal{B}^t$ give

$$1 + g + \ell < 5b\mathcal{B}^t.$$

The sparse contribution to J_c is less than

$$\frac{50M_{\mathrm{coef}}b^2\mathcal{B}^{M+1}}{\mathcal{B}-1}.$$

The bound $b \geq 4096M_{\mathrm{coef}}$ gives $50M_{\mathrm{coef}}b^2 < \mathcal{B}/64$. Adding the geometric mask therefore leaves $J_c < N_1$, and $0 < S_c, T_c < N$. This is the only changed numerical estimate in the inherited source argument; using the old numerator 32 after adding ℓ would not justify the broad-range bound.

If b is a power of two, the no-carry proof applies. It recovers digits $0 \leq z_i < b$ and positive witnesses $p_i = z_i + 1$. The coefficients to be centered are bounded by

$$M_{\mathrm{coef}}\left(1 + \sum_i p_i\right)^2 \leq M_{\mathrm{coef}}(32b)^2 \leq \mathcal{B}/4 < \mathcal{B}/2.$$

The independent target certificate supplies exact isolation, so the high mask forces all $r_j(\mathbf{p}) = 0$. Conversely a positive parent zero permits an arbitrarily large binary $b \geq b_{\mathrm{floor}}$ and $g = \sum_i (p_i - 1)\mathcal{B}^{w_i} > 0$ with $g < b\mathcal{B}^t$. Thus the exact interface is

$$\text{parent zero} \implies \text{binary } b, \quad N^2 \mid \binom{2R}{R}, \quad 32g < G_{\mathrm{cap}},$$

$$\text{binary } b, \quad N^2 \mid \binom{2R}{R}, \quad 8g < G_{\mathrm{cap}} \implies \text{parent zero}.$$

8.2 The full finalizer

All notation in this subsection is local to the nine-variable branch. Define

$$\begin{aligned}
&U = N^2 w, && Y = N^2 s, && M_0 = RY, && P = 2M_0^2 U,\\
&K = R + 1 + h(P-1), && A = M_0(U+1), && B_1 = 2R+1, && W_1 = bw,\\
&C = B_1 + W_1 + \phi, && J_* = 4A - 5, && Q = 3W_1 C - 2(W_1^2 - 1), &&\\
&D = (A^2-1)C^2 + 1, && E = iJ_* DC^2, && F = (A^2-1)E^2 + 1, &&\\
&G = A + F(CD + 1 - A), && H = B_1 + jC, && I = (G^2-1)H^2 + 1, &&\\
&a_0 = (P^2-1)K^2 + 1, && a_1 = DFI, && u = J_* F, && v = QF + J_*(H - C).
\end{aligned}$$

The ratio margin and strict square-root majorants are

$$\begin{aligned}
&\mu = \big(K^2 - 4(C - KY)^2\big)G_{\mathrm{cap}} - 8gK^2,\\
&V_0 = PK, \qquad V_1 = (AC)(AE)(GH), \qquad M_{\mathrm{div}} = Q + H,\\
&\omega = u\mu, \qquad \theta = u\big(n + M_{\mathrm{div}} - \mu(2M_{\mathrm{div}} + V_0 V_1)\big) + v,\\
&b_0 = \omega^2 a_0, \qquad b_1 = \omega^2 V_0^2 a_1.
\end{aligned}$$

The equation is

$$\boxed{F_9 = (\theta^2 - b_0 - b_1)^2 - 4b_0 b_1 = 0.} \tag{15}$$

To obtain the strictly positive version, replace the mixed coordinates ϵ, ϕ, n by their respective positive coordinates minus one. To obtain the nonnegative version, retain those three coordinates and replace each of g, h, s, w, i, j by its nonnegative coordinate plus one. These are domain bijections. The equation is not asserted to have the same consistency property when its inputs are allowed to be arbitrary signed integers.

8.3 Why one equation enforces the ratio conditions

The inherited Jones ratio theorem has the following form. Under $R, N \geq 8$, $b \leq R, N$, and $b, G_{\mathrm{cap}} > 0$, the conditions

$$a_0, a_1 \text{ squares}, \qquad u \mid v, \qquad \mu > 0$$

imply binary b, $N^2 \mid \binom{2R}{R}$, and $8g < G_{\mathrm{cap}}$. Conversely, binary b, the divisibility, and $32g < G_{\mathrm{cap}}$ supply positive h, s, w, i, j and nonnegative ϕ with those conditions. The complete dependency identification and inherited finalizer are in `upstream/PACKAGE45_NINE_PROOF.md`. There are also existing Lean proof sources for `Jones1982.Nine.ratio_sound` and `Jones1982.Nine.ratio_complete` in `Diophantine.Paper1982.NineRatio`. Their signatures have arithmetic hypotheses, with no machine or compiler assumption. The archive includes their 20-file local import closure and a source-hash audit under `upstream/JONES_RATIO_LEAN/`. This is inspected existing proof source; a new Lean kernel replay was not performed, and Mathlib remains an external dependency.

For completeness we give the algebraic finalizer proof, since it distinguishes a valid domain translation from merely adding variables. On every mixed-domain assignment, all of $P, K, A, C, J_*, D, E, F, G, H$ exceed one, while $a_0, a_1, u > 0$. The exact identities

$$\begin{aligned} V_0^2 - a_0 &= K^2 - 1 > 0, \\ (AC)^2 - D &= C^2 - 1 > 0, \\ (AE)^2 - F &= E^2 - 1 > 0, \\ (GH)^2 - I &= H^2 - 1 > 0 \end{aligned}$$

give $0 < a_0 < V_0^2$ and $0 < a_1 < V_1^2$. Moreover $Q > 0$, $H - C > 0$, and

$$0 < \frac{v}{u} = \frac{Q}{J_*} + \frac{H - C}{F} < M_{\mathrm{div}} - 1.$$

If $\mu = 0$, then $\omega = 0$ and $\theta/u = n + M_{\mathrm{div}} + v/u > 0$, excluding a zero of (15).

Otherwise the four-factor norm identity gives $\theta/\omega = r_0 + V_0 r_1$ with $r_a^2 = a_a$. The roots are real. The rational sum cannot vanish, since that would give $|r_1| = |r_0|/V_0 < 1$, contradicting the positive integer a_1. Squaring the nonzero rational sum makes both roots rational and hence integers. The strict bounds yield $|r_a| \leq V_a - 1$, so

$$Z = 2M_{\mathrm{div}} + V_0 V_1 + r_0 + V_0 r_1 \geq 2M_{\mathrm{div}} + 1 > 0.$$

The vanishing factor now says

$$n + M_{\mathrm{div}} + v/u = \mu Z.$$

The left side is positive, giving $\mu > 0$. Its remaining terms are integers, giving $u \mid v$. Conversely, from the two squares, $u \mid v$, and $\mu > 0$, choose integer roots and set $n = \mu Z - M_{\mathrm{div}} - v/u$. The same bounds give $n \geq 2$, and a norm factor vanishes.

The source interface and the ratio theorem now prove that (15) has a zero exactly when the positive parent does. This proves the nine-variable cases of Theorem 1.1.

8.4 Exact degree despite cancellation

Put $d_N = 3(E_{\max} + 3t + 2) = 28\,926$. On the diagonal the source has N of degree d_N and leading coefficient 4, and R of degree $3d_N$ and leading coefficient 16. The new sparse translation is strictly lower degree than the geometric mask, so the inherited finalizer's degree ledger still applies.

Naive propagation in (15) would conceal a leading cancellation. Resolve it with the exact identity

$$\begin{aligned} V_1^2 - a_1 = (C^2 - 1)(AE)^2(GH)^2 \\ + D(E^2 - 1)(GH)^2 + DF(H^2 - 1). \end{aligned}$$

The first two summands each have degree $298d_N + 64$ and diagonal leading coefficient 2^{518}; the third has lower degree. With $E_0 = u(n + M_{\mathrm{div}}) + v - 2\omega M_{\mathrm{div}}$, exactly

$$\theta^2 - \omega^2(a_0 + V_0^2 a_1) = E_0^2 - 2E_0\omega V_0 V_1 + \omega^2 V_0^2(V_1^2 - a_1) - \omega^2 a_0.$$

The third term uniquely dominates, with degree $576d_N + 6t + 140$ and leading coefficient 2^{1025}. The final subtracted term in (15) has lower degree. Therefore

$$\deg F_9 = 1152d_N + 12t + 280 = 33\,332\,620, \qquad [T^{33\,332\,620}]F_9(T, \dots, T) = 2^{2050}.$$

The coordinate shifts preserve this leading term. The primary positive evaluator uses 682 operations; its natural companion uses 685. Separate 700/703-operation circuits make the degree cancellation explicit. Their greater costs are not substituted for the primary evaluator counts.

9 A squared-modulus quotient: a 66-operation equation

The constructions with fewest variables need large arithmetic circuits. There is a separate improvement at the short-circuit end of the comparison: the inherited fifteen-positive-variable equation can be evaluated in **66 binary arithmetic operations**. Its exact degree is 157. This saves one operation against the repository's 67-operation, degree-155 baseline. The same idea lowers the costs of the degree-80 and degree-68 alternatives from 73 and 75 to 72 and 74 operations, respectively. The saving comes from a stronger auxiliary congruence, followed by a change of the quantified quotient. It is not an identity between the old and new polynomials at the same quotient coordinate.

9.1 The explicit polynomial and its fixed coefficients

Use the entire coefficient tuple of package 36 at the source commit `124c07c0b7`: the ports are $\beta = B - 1, K_0, M_C, M_F$, with $B = 2^{1012115}$. Here K_0 is the tuple's `K` and M_F is its already shifted `MF_source`. The four ports are fixed integers, not additional unknowns. The coefficient construction and its machine semantics are inherited; the argument below introduces no rebinding or change of masks.

All fifteen unknowns are strictly positive integers:

$$(J, F, \alpha, z, f, h, i, T, s, w, \tau, \eta, \zeta, y, \gamma) \in \mathbb{Z}_{>0}^{15}.$$

Define the following polynomial abbreviations, which introduce no new unknowns:

$$\begin{aligned}
q &= \beta J + 1, & C &= q - F - \alpha, & & \\
X &= wq, & Y &= sq^3, & E &= XY, \\
k &= \eta + \zeta, & c &= kY + \eta, & a &= E + Y, \\
H &= 4a + 3, & \Delta &= a^2 + H, & A &= a + 2, \\
D &= X + ac + \gamma H, & G &= (q-1)(q-F) + \alpha, & & \\
R &= G(q^2 - 1) + (M_C + qM_F)J, & S &= i\Delta c^2, & & \\
V &= f^2(cT - R) - c. & & & &
\end{aligned}$$

In particular, $\Delta = A^2 - 1 = (a+1)(a+3) > 0$ on the positive domain. Set

$$\begin{aligned}
N_0 &= \tau^2 - (XY^2)(XY^2 + 1)k^2, \\
N_1 &= D^2 - \Delta c^2, \\
N_a &= S^2(V^2 - y^2) + y^2, \\
N_k &= k - hE - R, \\
N_t &= (K_0 + w)C + (q - F) - z(q-1), \\
N_{\mathrm{str}} &= \Delta f^2 - S^2.
\end{aligned}$$

The proposed equation is the ordinary integer polynomial equation

$$\boxed{P_{66} := N_0 N_1 N_a N_k N_t N_{\mathrm{str}} - \Delta = 0.} \tag{16}$$

Every exponent in this definition is fixed. The divisions and Pell polynomials used below occur only in the proof and witness construction.

9.2 Why a square modulus permits a cheaper quotient

Write $\chi_t(n), \psi_t(n)$ for the integer Pell polynomials determined by

$$\chi_t(0) = 1, \qquad \psi_t(0) = 0,$$
$$\chi_t(n+1) = t\chi_t(n) + (t^2-1)\psi_t(n), \qquad \psi_t(n+1) = \chi_t(n) + t\psi_t(n).$$

Thus $\chi_t(n)^2 - (t^2-1)\psi_t(n)^2 = 1$.

Lemma 9.1 (Odd Pell quotient identities)**.** *For each $r \in \mathbb{N}$ there is a polynomial $Q_r \in \mathbb{Z}[Z]$ such that*

$$\chi_t(2r+1) = tQ_r(t^2), \tag{17}$$
$$Q_r(0) = (-1)^r(2r+1), \tag{18}$$
$$Q_r(1-A^2) = (-1)^r\psi_A(2r+1). \tag{19}$$

Proof. The odd-index subsequence gives

$$Q_0(Z) = 1, \qquad Q_1(Z) = 4Z - 3, \qquad Q_{r+1}(Z) = (4Z-2)Q_r(Z) - Q_{r-1}(Z).$$

This proves (17) and (18) by induction. The sequence $\psi_A(2r+1)$ starts with $1, 4A^2-1$ and has recurrence coefficient $4A^2-2$. Substituting $Z = 1-A^2$ changes the coefficient of the displayed recurrence to its negative, giving (19) by the same induction. These are identities over the integers and therefore also hold modulo any positive integer. □

Lemma 9.2 (Squared-modulus reconstruction)**.** *Suppose $A > 1$, $R > 0$, $R \equiv 3 \pmod 4$, $c = \psi_A(R) > 0$, and $i, f > 0$ satisfy*

$$\Delta = A^2 - 1, \qquad f^2 = 1 + \Delta i^2 c^4.$$

Put $S = \Delta i c^2$. There are positive integers V, y, T satisfying

$$S^2(V^2 - y^2) + y^2 = 1, \qquad V = f^2(cT - R) - c.$$

Proof. Take

$$V = \frac{\chi_S(R)}{S}, \qquad y = \psi_S(R).$$

The quotient is integral by (17); both values are positive because $S > 1$ and $R > 0$. The Pell identity gives the required norm equation. Also

$$S^2 = \Delta(f^2 - 1) \equiv 1 - A^2 \pmod{f^2}.$$

Since $(R-1)/2$ is odd, Lemma 9.1 gives the two congruences

$$V \equiv -c \pmod{f^2}, \qquad V \equiv -R \pmod{c}. \tag{20}$$

For the second, use $c \mid S$ and (18). The strong equation gives $f^2 \equiv 1 \pmod c$, hence $\gcd(c, f) = 1$. The numerator $V + c + Rf^2$ is therefore divisible by both f^2 and c, and thus by cf^2. Consequently

$$T = \frac{V + c + Rf^2}{cf^2}$$

is a positive integer. Rearranging proves the stated quotient identity. □

The earlier auxiliary construction used congruence modulo f. Retaining the full modulus f^2 is what makes the old quotient divisible by f, after choosing a suitable auxiliary Pell ordinate.

9.3 Soundness, completeness, and the native congruence

Theorem 9.3 (Existence-equivalence of the short equation)**.** *With the authentic package-36 coefficient tuple, the inherited ordinary mathematical compiler and K215 calibration imply*

$$\mathrm{ZFC} \vdash \left[\neg\exists \mathbf{x} \in \mathbb{Z}_{>0}^{15}\ P_{66}(\mathbf{x}) = 0\right] \iff \mathrm{Con}(\mathrm{ZFC}).$$

The additional reduction from the inherited explicit polynomial to (16) *requires no consistency assumption.*

Proof. Let P_{67} denote the inherited polynomial on the same fifteen slots. Its only different abbreviation is

$$V_{\mathrm{old}} = c(T_{\mathrm{old}} f - 1) - Rf^2.$$

The exact all-value identity

$$P_{66}(\ldots, f, T, \ldots) = P_{67}(\ldots, f, fT, \ldots) \tag{21}$$

follows by substitution. Thus every positive new zero gives a positive old zero, since $fT > 0$. Inherited soundness supplies a finite ZFC refutation.

Conversely, a finite refutation supplies a positive old zero. Its inherited Pell and compiler recovery gives

$$c = \psi_A(R), \qquad f^2 = 1 + \Delta i^2 c^4, \qquad q = B^N, \qquad J = 1 + B + \cdots + B^{N-1}$$

for some positive integer N, together with the native bit condition

$$(C - 1)\ \mathrm{AND}\ (M_C J + 1) = 0.$$

The tuple has $4 \mid B$ and $M_C \equiv 2 \pmod 4$. Hence $q \equiv 0 \pmod 4$, $J \equiv 1 \pmod 4$, and $M_C J + 1 \equiv 3 \pmod 4$. The two low bits of the nonnegative integer $C - 1$ must vanish, so $C \equiv 1 \pmod 4$. Using the exact identity $G = q(q - F) - C$, the definition of R yields

$$R \equiv C + 2 \equiv 3 \pmod 4.$$

This phase follows from the authentic native decoding; it is not a restriction silently imposed on arbitrary coefficient assignments.

Apply Lemma 9.2 at these same values of A, R, c, i, f. Replace only the supplied auxiliary ordinate y and quotient T by the lemma's positive integers. The auxiliary norm factor is again one. The other factors are unchanged; in particular the inherited unit classification gives $N_0 = N_1 = N_k = N_t = 1$ and $N_{\mathrm{str}} = \Delta$. Thus $P_{66} = 0$.

We have established equivalence of positive solvability, which composes with the inherited refutation calibration. Negating gives the displayed consistency equivalence. All new steps are finite integer algebra, induction on polynomial recurrences, and elementary divisibility, and can be carried out in ZFC. □

The reverse map may change both auxiliary coordinates. In particular, the theorem does not assert that every old quotient is divisible by f, or that the old and new equations have identical zero sets at unchanged coordinates. The local substitution proof also does not supply an end-to-end Lean or Rocq certificate for the inherited native-window compiler.

9.4 Exact arithmetic size, degree, and companion equations

The old evaluator already computes f^2. Its five private quotient operations are replaced by the four operations

$$cT, \qquad cT - R, \qquad f^2(cT - R), \qquad f^2(cT - R) - c.$$

Every other gate is retained. The complete new evaluator has 38 multiplications, 15 additions, and 13 subtractions. Multiplication by a fixed numeral is charged; reuse is free. All gates, all fifteen witnesses, and all four coefficient ports are live.

The exact factor degrees in the order $N_0, N_1, N_a, N_k, N_t, N_{\mathrm{str}}$ are

$$22, \quad 18, \quad 62, \quad 7, \quad 2, \quad 46.$$

Put $Q = \beta J$, $k_0 = \eta + \zeta$, and $C_1 = Q - F - \alpha$. The leading homogeneous polynomial of P_{66} is

$$-8Q^{91}h\gamma i^4 k_0^{13} w^{13} s^{26} T^2 f^4 (wC_1 - zQ).$$

In particular, the coefficient of

$$J^{92} h\gamma i^4 \eta^{13} w^{13} s^{26} T^2 f^4 z$$

is $8\beta^{92} \neq 0$. The exact degree is therefore 157, and the final subtraction of Δ, of degree 12, cannot affect it.

For lower-degree alternatives, retain the new V but put

$$\widetilde{N}_a = \Delta(f^2 - 1)(V^2 - y^2) + y^2, \qquad N_s = f^2 - \Delta(ic^2)^2.$$

The following three companion equations inherit the same positive solvability equivalence:

$$\begin{aligned}
P_{68} &= N_0 N_1 \widetilde{N}_a N_k N_t N_s - 1 = 0, \\
P_{72} &= (N_0 N_1 - 1)^2 + (\widetilde{N}_a N_k N_t - 1)^2 + (N_s - 1)^2 = 0, \\
P_{74} &= (N_0 - 1)^2 + (\widetilde{N}_a - 1)^2 + (N_s - 1)^2 + (N_1 N_k N_t - 1)^2 = 0.
\end{aligned}$$

Indeed each is the corresponding inherited parent equation under the same positive substitution $T_{\mathrm{old}} = fT$; the established parent equivalences and the reconstruction proof above apply. Their fully paid counts and exact degrees are

	multiplications	additions	subtractions	total	degree
P_{66}	38	15	13	66	157
P_{68}	39	15	14	68	113
P_{72}	40	17	15	72	80
P_{74}	40	18	16	74	68

For P_{72}, the first squared residual uniquely reaches degree 80; its leading form is $64\gamma^2 w^8 s^{14} (\eta+\zeta)^6 (\beta J)^{50}$. For P_{74}, the strong residual uniquely reaches degree 68 after squaring; its leading form is $i^4 w^4 s^{12} (\eta + \zeta)^8 (\beta J)^{40}$. For P_{68}, a degree-113 monomial has coefficient $8\beta^{64}$. These nonzero coefficients certify the claimed exact degrees uniformly for the prescribed tuple.

The supplied standard-library auditor checks the literal schedules, source hashes, exact polynomial identities at the changed cut, leading homogeneous forms, and full liveness. Normal and optimized Python executions agree. Its finite congruence examples and modular circuit diagnostics support the implementation; the universal reconstruction lemma, rather than those examples, supplies the new mathematical proof. The result improves stated repository tradeoffs and makes no claim of global minimality.

10 Reproduction, certificates, and proof status

10.1 What is fully explicit

The archive supplies ordinary integer polynomials as finite, acyclic, division-free arithmetic circuits. The data distinguishes input variables, fixed integer leaves, and binary arithmetic nodes. All outputs depend

on the advertised variables, and every counted gate lies on the output's dependency graph. There is no hidden existential parameter in a coefficient.

Both complete native tuples are included with their deterministic compilers and literal machine/tile data. The low-variable equations use the coherent package-31 tuple; the fifteen-variable equations use the coherent package-36 tuple. Derived fixed leaves are reconstructed from explicit integer recipes and checked against their payloads. Mathematical abbreviations such as $\psi_A(R)$ occur only in proofs, never as polynomial gates.

The large fixed constants matter when interpreting simplicity. These improvements concern the number of witnesses, total degree, and the stated arithmetic-circuit cost. They do not claim a decrease in every measure of description length or coefficient height. Expanding the complete polynomials into lists of all monomials is unnecessary and would obscure their exact finite definitions.

10.2 Independent finite checks

The supplied validators establish the following implementation facts.

1. **Exact source binding.** An independently written sparse symbolic expansion agrees with the parent arithmetic DAG over the polynomial ring in the native coefficient ports. All 528 padded quadratic positions are distinct. The 94 reversal records satisfy all 1,974 off-diagonal conditions. Enumeration of all 49,632 convolution pairs gives exactly the intended target coefficients, including terms absent from individual residuals.

2. **Literal finalizer replay.** Independent evaluators reproduce the full ten-variable formula and the nine-variable formula, coordinate translations, and cancellation-resolved certificate. These are implementation diagnostics; their finite samples are not represented as universal identities.

3. **Exact algebra and degrees.** The three-root norm identity is compared symbolically as a 35-term polynomial. The nine-variable telescoping identity is checked exactly. Degree upper bounds and nonzero leading coefficients prove the reported degrees. The short circuits have explicit exact cut identities and leading homogeneous forms.

4. **Guard and toy checks.** The separator audit includes 157,216 coprimality cases, 22,680 parity cases, 7,750 Pell-remainder cases, and four explicit counterexamples to dropping needed guards. Small accepting and rejecting carry-mask examples exercise the source in nonvacuous situations. These tests supplement the universal proofs above.

5. **Reconstruction and robust execution.** The release checker rebuilds in a temporary copy and compares protected artifacts byte for byte. Assertions needed for correctness are explicit checks and remain active under Python's optimization flag.

Run from the extracted archive:

```
python3 verify_all.py
```

The mathematical verifiers use the Python standard library. Building the article additionally requires a usual LaTeX installation:

```
cd report
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

The README provides the exact file map and separate commands.

10.3 What these certificates do and do not establish

There are three distinct levels of justification. First, the new selective-extraction, packed-successor, arithmetic-separator, and squared-modulus results are proved as ordinary integer mathematics in this report and its detailed companion proofs. Second, the exact delivered circuits, coefficients, and degree/cost claims have reproducible finite certificates. Third, the semantic endpoint imports the repository's native-window compiler theorem, fixed K215 calibration, and the existing Lucas/ratio theorems identified in the provenance records.

The complete new chain has not been rebuilt in Lean or Rocq. A finite circuit audit does not certify a machine's unbounded semantics, and a randomized replay does not prove an algebraic identity. The existing active Lean development proves several inherited local identities and formal existence theorems; those are not silently promoted to a kernel proof of the new explicit native-compiler composition.

All new reasoning uses finite algebra, elementary number theory, and standard arithmetic theorems formalizable in ZFC. Subject to the stated inherited mathematical theorems, the composition proves

$$\mathsf{ZFC} \vdash (\neg\exists \mathbf{x} \in D\ F(\mathbf{x}) = 0) \longleftrightarrow \mathrm{Con}(\mathsf{ZFC})$$

for the specified domains D. This equivalence does not decide which side holds. In particular, no experimental search for witnesses or assumption of consistency was used to manufacture the coefficients.

11 What changed, and promising next questions

The improvements come from three structural changes and one separate Pell refinement.

Use the support of each residual. Uniform block spacing protects many positions that no residual can use. The new target rule forbids only actual cross-interference against the full dense quadratic square. It reduces the maximum target from 35 177 to 7 243 and the packed span from 36 775 to 8 841. The target list is a finite certificate; the heuristic that found it is not part of the proof.

Translate positivity after packing. The identity $1 + \sum_i (n_i + 1)Z^{w_i} = 1 + \sum_i n_i Z^{w_i} + \ell(Z)$ avoids expanding a successor substitution separately in every residual. Weighted records drop from 136 to 94 and port-expanded terms from 161 to 99. In these evaluators, the number of nonliteral fixed leaves drops from 17 to 10. The small change of coefficient representation also creates more freedom for target placement.

Separate radicals using arithmetic. The previous weights grow enough to dominate earlier square roots. The new weights $(1, b, bK)$ exploit a coprime factorization, the modulo-four structure of the last two radicands, and the constant remainder in a Pell equation. This removes a large source of degree without adding a quantified variable. The unit case $b = 1$ is essential: ignoring it would leave a gap exactly where the positivity register vanishes.

Keep the full square modulus in the short equation. The stronger congruence modulo f^2 lets the reconstructed old quotient be a multiple of f. Reparameterizing that quotient removes one paid gate. This is a change in existential presentation, so its completeness proof must reconstruct witnesses; a same-coordinate symbolic rewrite alone would not suffice.

The highest-value next targets are consequently specific. One is to jointly optimize the positive-parent presentation, Sidon positions, target placement, and addition chains, with exact certificates kept separate from search heuristics. Another is to classify when a Pell remainder can replace a general magnitude separator in other relation-combining constructions. A third is to formalize the source interface and new

separator as reusable lemmas, then connect them to a fully checked native-compiler endpoint. A lower variable count would require an additional structural idea; none is claimed by the present work.

Attribution and integration notes

The literal K215 machine, native-window compiler, original parents, coefficient tuples, quintic merger, and earlier finalizers belong to the cited ProveIt development and its credited mathematical sources. Sun's ten-integer-variable theorem is not a new result of this report. The nine-positive branch retains the Jones ratio theorem and the package-45 strict-majorant finalizer.

The new contributions here are the support-sensitive target theorem and certified schedule, the packed successor translation, the arithmetic separator and its signed bootstrap, the squared-modulus quotient reconstruction, and their explicit compiled combinations. Exact gate sharing is identified separately from these mathematical changes.

The package is intended as a new research contribution beside the pinned existing packages. It leaves those sources unchanged. A suggested directory name is `46-sparse-arithmetic`; repository maintainers can choose the final numbering. Preserve the included upstream attribution and licensing notices when integrating copied coefficient and proof artifacts.

References

[1] V. Reshetnikov and contributors, *ProveIt*, source snapshot `124c07c0b77a9e20e84676514d8b1ff78536ea3f`, reviewed 23 October 2026. https://github.com/VladimirReshetnikov/ProveIt. The complete file-level source pins are included in the archive.

[2] ProveIt, *40-packing10*, in `Computability/DiophantineConsistency/con-zfc-diophantine-forms`, at the source snapshot above. The report, source parent, inherited finalizer proof, and notices are bundled under `upstream/`.

[3] ProveIt, *45-multioutput*, at the same source snapshot. Its nine-variable proof and dependency attribution are bundled as `upstream/PACKAGE45_NINE_PROOF.md`.

[4] ProveIt, *36-small-tuple*, at the same source snapshot. The full coefficient data and inherited P67 proof are bundled with the short-equation contribution.

[5] Z.-W. Sun, *Further results on Hilbert's Tenth Problem*, Science China Mathematics **64** (2021), 281–306. https://arxiv.org/abs/1704.03504.

[6] Z.-W. Sun, *Ten unknowns for Hilbert's tenth problem over the integers*, arXiv:2609.01594v2 (2026). https://arxiv.org/abs/2609.01594v2. The ten-integer composition uses the modified Lucas certificate, residue interpolation, and binomial bridge, as identified in the included inherited proof.